

Cybersecurity Training Diary

Day 5: Common Cyber Attacks

1. Topics Covered

✓ Malware Attacks

Type	Description	Example	Prevention
Virus	Self-replicating, needs host file	ILOVEYOU	Antivirus, file scanning
Worm	Spreads autonomously	WannaCry	Patch management
Trojan	Disguised as legitimate	Zeus	User education
Ransomware	Encrypts files for ransom	NotPetya	Regular backups

✓ Social Engineering Attacks

- **Phishing:** Fake emails (e.g., "Your account is locked!")
- **Speare Phishing:** Targeted phishing (e.g., CEO fraud)
- **Vishing:** Voice call scams ("Tech support" calls)
- **Smishing:** SMS phishing ("Click this delivery link")

✓ Network Attacks

- **DDoS:** Overwhelming traffic (Mirai botnet)
- **Man-in-the-Middle:** Eavesdropping on connections
- **DNS Spoofing:** Redirecting to malicious sites

✓ Web Application Attacks

- **SQL Injection:** ' OR 1=1 -- in login forms
- **XSS:** Injecting scripts into webpages
- **CSRF:** Forcing authenticated actions

2. Key Takeaways

95% of breaches involve human error (Verizon DBIR)

Ransomware attacks occur every 11 seconds (2023 stat)

Patch management prevents 60% of attacks

3. Practical Exercises

Exercise 1: Analyze Phishing Email

1. Download a sample phishing email
2. Identify red flags:
 - Suspicious sender (admin@amaz0n.com)
 - Urgent action required
 - Mismatched links (hover before clicking)

Exercise 2: Simulate SQL Injection

Use DVWA (Damn Vulnerable Web App):

```
' UNION SELECT username, password FROM users --
```

4. Questions for Reflection

Why do phishing attacks still work in 2024?

How would you explain ransomware risks to a non-technical team?

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